

# Retrieval-Augmented Generation for Domain-Specific Customer Support: Chunking Strategies, Embedding Models, and Latency-Accuracy Trade-offs

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## Abstract

Large Language Models (LLMs) exhibit impressive generative capability but suffer from hallucination and knowledge staleness when deployed in enterprise customer-support contexts. Retrieval-Augmented Generation (RAG) mitigates these issues by grounding LLM responses in a dynamically retrieved knowledge base. However, the quality of retrieval is highly sensitive to how documents are split into chunks and which embedding model converts them to dense vectors. This paper benchmarks four chunking strategies — fixed-size, sentence-boundary, semantic, and hierarchical — against three embedding models (OpenAI text-embedding-3-small, BGE-M3, and multilingual-e5-large) on a Vietnamese e-commerce customer-support dataset containing 12 000 FAQ entries and 3 500 annotated conversation turns. We evaluate each combination on Recall@5, Mean Reciprocal Rank (MRR), end-to-end answer faithfulness, and p95 response latency. Semantic chunking paired with BGE-M3 embeddings achieves the best Recall@5 (0.91) and faithfulness (0.88), while fixed-size chunking with multilingual-e5-large offers the best latency-accuracy trade-off for latency-sensitive deployments.

**Keywords:** retrieval-augmented generation, RAG, LLM, chunking strategy, embeddings, customer support, Vietnamese NLP, vector search

## 1. Introduction

Enterprise customer-support applications present a demanding evaluation environment for LLMs. Users expect accurate, up-to-date answers about product specifications, return policies, and order statuses — information that changes frequently and lies outside any pre-training corpus. Moreover, Vietnamese — the primary language of our target deployment — is morphologically richer than English and under-represented in most open-source pre-training datasets, making pure prompt-based approaches particularly unreliable.

RAG addresses these challenges by splitting the task into two stages: (1) a retriever fetches the  $k$  most relevant document chunks from a vector store given the user query, and (2) an LLM generates a response conditioned on the retrieved context. The quality of the final answer is thus bounded by retrieval quality, which in turn depends on chunk granularity and the fidelity of the embedding space.

Despite the practical importance of these design decisions, comparative studies on non-English, domain-specific corpora remain scarce. This paper fills that gap with a systematic benchmark across four chunking strategies and three embedding models on a proprietary Vietnamese e-commerce dataset.

## 2. Related Work

Lewis et al. (2020) introduced the RAG framework, demonstrating that retrieval-augmented models outperform closed-book LLMs on knowledge-intensive NLP tasks. Subsequent work explored dense passage retrieval (DPR), ColBERT-style late interaction, and hybrid sparse-dense retrieval. Chunking has received less systematic attention; most practitioners use fixed-size windows of 256–512 tokens as a default. Kamradt (2023) proposed semantic chunking using cosine similarity between consecutive sentence embeddings to identify topical boundaries, but did not evaluate it against non-English corpora.

For Vietnamese NLP, PhoBERT and multilingual models such as mBERT and XLM-RoBERTa provide reasonable representation, but embedding-specific evaluations on retrieval tasks are limited. Our work is the first to compare BGE-M3 — a recent multilingual embedding model with demonstrated strong cross-lingual performance — against multilingual-e5-large on a Vietnamese domain-specific retrieval benchmark.

## 3. Methodology

### 3.1 Dataset

The dataset consists of 12 000 FAQ question-answer pairs from a Vietnamese e-commerce platform and 3 500 annotated support-conversation turns where the correct source document chunk is labelled. FAQ entries range from 40 to 800 tokens. All data was collected with user consent and anonymised before use.

### 3.2 Chunking Strategies

| Strategy          | Description                                                          | Avg. Chunk Size |
|-------------------|----------------------------------------------------------------------|-----------------|
| Fixed-size        | Split every N tokens with M-token overlap                            | 256 tokens      |
| Sentence-boundary | Split at sentence boundaries, merge until size limit                 | 180 tokens      |
| Semantic          | Split when cosine similarity between sentences drops below threshold | 210 tokens      |
| Hierarchical      | Two-level: paragraph chunks with sentence sub-chunks                 | 320 / 90 tokens |

Table 1. Chunking strategies evaluated in this study.

### 3.3 Embedding Models

We evaluate: (a) OpenAI text-embedding-3-small (1 536 dimensions, API-based); (b) BAAI/bge-m3 (1 024 dimensions, local inference via sentence-transformers); (c) intfloat/multilingual-e5-large (1 024 dimensions, local inference). All models are used with their default instruction prefixes where

applicable.

### 3.4 Evaluation Metrics

We report Recall@5 (fraction of queries for which the correct chunk appears in the top-5 results), Mean Reciprocal Rank (MRR), Answer Faithfulness (GPT-4o-mini judged, 0–1 scale), and p95 end-to-end latency measured on a single A10G GPU node for local models.

## 4. Results

| Chunking          | Embedding Model        | Recall@5 | MRR  | Faithfulness | p95 Latency |
|-------------------|------------------------|----------|------|--------------|-------------|
| Fixed-size        | text-embedding-3-small | 0.81     | 0.74 | 0.79         | 320 ms      |
| Fixed-size        | BGE-M3                 | 0.83     | 0.76 | 0.81         | 480 ms      |
| Fixed-size        | multilingual-e5-large  | 0.82     | 0.75 | 0.80         | 390 ms      |
| Sentence-boundary | text-embedding-3-small | 0.85     | 0.78 | 0.83         | 330 ms      |
| Sentence-boundary | BGE-M3                 | 0.87     | 0.80 | 0.85         | 490 ms      |
| Semantic          | BGE-M3                 | 0.91     | 0.85 | 0.88         | 520 ms      |
| Semantic          | multilingual-e5-large  | 0.88     | 0.82 | 0.85         | 410 ms      |
| Hierarchical      | BGE-M3                 | 0.89     | 0.83 | 0.86         | 640 ms      |

Table 2. Benchmark results across chunking strategies and embedding models.

Semantic chunking with BGE-M3 achieves the best Recall@5 (0.91) and faithfulness (0.88), confirming that preserving topical coherence within chunks benefits dense retrieval. The improvement over fixed-size chunking with the same model is statistically significant (paired t-test,  $p < 0.01$ ). Hierarchical chunking yields high recall but the highest latency due to the two-stage retrieval process.

## 5. Analysis and Discussion

BGE-M3’s advantage over multilingual-e5-large is most pronounced on Vietnamese-specific queries involving compound nouns and colloquial abbreviations common in chat-based customer support (e.g., ‘ship COD’, ‘hoàn ti■n’). We attribute this to BGE-M3’s larger and more diverse multilingual pre-training corpus.

For latency-sensitive deployments where p95 must remain below 450 ms, fixed-size chunking with multilingual-e5-large provides the best accuracy-latency balance (Recall@5 = 0.82, p95 = 390 ms). Operators can further reduce latency by serving the embedding model with TensorRT-LLM quantised to INT8, which we measured to reduce inference time by 38% with negligible accuracy degradation.

A qualitative error analysis of the 9% of queries where semantic chunking with BGE-M3 failed to retrieve the correct chunk revealed three dominant failure modes: (1) the relevant information was spread across multiple FAQ entries with no single chunk containing a complete answer; (2) the user query used regional dialect terms not present in the FAQ; (3) the query was ambiguous without conversational history.

## 6. Conclusion

We presented a systematic benchmark of RAG pipeline components for Vietnamese e-commerce customer support. Semantic chunking paired with BGE-M3 embeddings achieves the best retrieval quality (Recall@5 = 0.91, faithfulness = 0.88). For deployments with strict latency budgets, fixed-size chunking with multilingual-e5-large is recommended. Future work will explore conversational RAG with multi-turn context fusion and query expansion using a smaller generative model before retrieval.

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